Unit 1 Lab: Easter Egg Hunt Edition

Description

In this lab, you will create a Python program that simulates an Easter egg hunt at Celebration Square. You will use for loops, while loops, variables, and conditionals to track eggs, celebrate finds, and calculate prizes.

This lab assesses your understanding of loops and how they work with variables and conditions.

Choose one of three challenge tiers: Bronze, Silver, or Gold. Each tier builds on the previous one.

Submission Instructions

Submit your .py file as unit1_lab_tier.py (e.g., unit1_lab_gold.py) via GitHub.

Bronze Tier — Max Grade of 70%

Tasks:

- Use print() to greet the user:
  - Welcome to the Celebration Square Easter Egg Hunt!
- Use input() to ask for the user's name and how many eggs they found today (integer).
- Print a message using those variables.
- Use a for loop to celebrate each egg one by one:
  - For every egg except the last: print "Egg X found! Keep going!"
  - For the last egg: print "Egg X found! Happy Easter!"
- Use input() to ask for the user's favourite Easter egg colour.

Sample Output:

```
Welcome to the Celebration Square Easter Egg Hunt!
What is your name? Alex
How many eggs did you find today? 3
Great job Alex! Let's celebrate each egg you found!
Egg 1 found! Keep going!
Egg 2 found! Keep going!
Egg 3 found! Happy Easter!
```

Silver Tier — Max Grade of 85%

Tasks:

- Complete all Bronze tasks.
- Print: "\nTime to log your egg basket!"
- User will enter each egg colour one at a time and type 'done' when finished.
- Use a while loop with the sentinel value "done" to collect egg colours: Keep a counter for total eggs AND golden eggs.
- Inside the loop:
  - If colour is "golden" → print "A golden egg! That's very rare!" and add to golden count
  - Otherwise → print "A {colour} egg added to your basket!"
- After the loop, print a basket summary:
  - Total eggs collected: X
  - Golden eggs found: X
- Use a conditional:
  - If golden count > 0 → "Amazing, {name}! You found at least one golden egg!"
  - Otherwise → "No golden eggs this time, {name}. Keep hunting!"

Sample Output for Silver:

```
Welcome to the Celebration Square Easter Egg Hunt!
What is your name? Alex
How many eggs did you find today? 3
Great job Alex! Let's celebrate each egg you found!
Egg 1 found! Keep going!
Egg 2 found! Keep going!
Egg 3 found! Happy Easter!

Time to log your egg basket!
Enter each egg colour (or 'done' to finish): red
A red egg added to your basket!
Enter each egg colour (or 'done' to finish): golden
A golden egg! That's very rare!
Enter each egg colour (or 'done' to finish): done
Total eggs: 2
Golden eggs: 1
Amazing, Alex! You found at least one golden egg!
```

Gold Tier — Max Grade of 100%

Concepts assessed: Everything in Bronze + Silver + point accumulation, second for loop

Note: You are ON YOUR OWN for programming the Gold conditions!

Tasks:

- Complete all Bronze and Silver tasks.
- MODIFY the loop from silver tier to track points.
- Update the inside the loop:
  - First, VALIDATE the input by only accepting “golden”, “speckled”, or “regular”. Keep asking the user for input until they type one of those three options. Note: capitalized words are valid inputs.
  - If colour is "golden" → print "A golden egg! That's very rare!", add to the golden count AND increase the points by 10
  - If colour is “speckled” → print “A speckled egg! You must be lucky!”, add to the speckled counter AND increase the points by 5
  - If colour is “regular” → print “A regular egg! Good job!”, add to the regular egg counter AND increase the points by 1
- After the loop, print:
  - Special Basket Summary:
  - Total eggs: X
  - Golden eggs: X
  - Speckled eggs: X
  - Regular eggs: X
  - Your total score: X points
- Print the prize tier based on total score:
  - Less than 10 → "Better luck next time!"
  - 11-15 → "Bronze Basket winner!"
  - 16-20 → "Silver Basket winner!"
  - 21 or more → "Gold Basket Champion!"

Sample Output for Gold:

```
Welcome to the Celebration Square Easter Egg Hunt!
What is your name? Alex
How many eggs did you find today? 3
Great job Alex! Let's celebrate each egg you found!
Egg 1 found! Keep going!
Egg 2 found! Keep going!
Egg 3 found! Happy Easter!
```

```
Time to log your egg basket!
Enter each egg colour (or 'done' to finish): golden
A golden egg! That's very rare!
Enter each egg colour (or 'done' to finish): regular
A regular egg! Good job!
Enter each egg colour (or 'done' to finish): blue
Invalid input.
Enter each egg colour (or 'done' to finish): speckled
A speckled egg! You must be lucky!
Total eggs: 3
Golden eggs: 1
Speckled eggs: 1
Regular eggs: 1
Your total score: 16
Amazing, Alex! You found at least one golden egg!
Silver Basket winner!
```

Reminders

- Your while loop MUST use a sentinel value to stop.
- Do not forget header comments with your name!
- Include IPO comments in the correct places (Input, Processing, Output).
- Start with Bronze, then build up to Silver and Gold if you have time.